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Analyzing the Effect of Evidence Quality in a Science-Domain RAG System

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Abstract

In this project, we conducted a controlled study to evaluate the impact of hyperparameters and document quality on a RAG systems performance using the SciQ dataset. We set up a normal RAG baseline configured with fixed parameters: a chunk size of 256 tokens, a retrieval volume of Top-k=3, and a standard embedding model all-MiniLM-L6-v2. To isolate the system's sensitivity to context quality, we compared this baseline against three configurations: LLM-only, oracle gold support, and corrupted documents. Our results demonstrated that RAG performance is constrained by evidence quality. The normal RAG performance fell between the oracle upper bound and the corrupted lower bound. The difference between the oracle and the RAG baseline pointed to an inherent error in the retrieval mechanism. We concluded that the exact answer sentence is the main driver of the effectiveness of the RAG system, since removing it causes the performance to drop to the almost LLM-only baseline level.

1 Introduction

Large Language Models (LLMs) are incredibly powerful, but they consistently struggle with highly specialized tasks because their knowledge is fixed in the training data and updating them takes a massive amount of computational cost. As a result, they frequently hallucinate and generate answers that sound confident and fluent, but are factually incorrect.

(Gao et al., 2023) explain that RAG offers a practical solution to this problem. Instead of forcing the LLM to memorize every single fact, RAG treats the task like an open-book exam. It dynamically fetches verified information from an external database to ground the LLM's response. This effectively bypasses the need for constant, expensive model retraining.

A standard RAG system consists of three main stages: document ingestion, retrieval, and generation. First, external documents are divided into smaller chunks and encoded as embeddings for storage in a vector database. Next, the retriever selects the most relevant chunks for a given query. Finally, the generator uses both the question and the retrieved evidence to produce a grounded answer.

2 Methodologies

2.1 Dataset

For this project, we used the science-domain QA dataset SciQ (Welbl et al., 2017). The dataset contains science exam questions and each question is presented with one correct answer and three distractor options.

A key advantage of SciQ for this project is that it provides an explicit gold supporting paragraph containing the evidence for the correct answer. This allows us to construct highly controlled upper- and lower-bound conditions.

2.2 System Pipeline

For our end-to-end baseline, we implemented a standard RAG pipeline consisting of FAISS for vector indexing, all-MiniLM-L6-v2 for text embeddings, and google/flan-t5-small for answer generation (Lewis et al., 2021). Documents are chunked into 256-token segments, and the RAG pipeline retrieves the top three most relevant chunks (k=3) and feeds them to the LLM to generate an answer. During the generation phase, we used a strict prompt template instructing the LLM to output only one option letter.

2.3 Main Experiments Design: Evidence Quality Bounds

To evaluate how much RAG performance is affected by evidence quality, we designed an experiment with four controlled conditions:

2.3.1 Baseline: LLM-only

The LLM answers each question without any retrieved or provided support document. This baseline helps measure how much external evidence contributes beyond the model's internal knowledge.

2.3.2 Normal RAG

This condition represents the realistic end-to-end RAG setting. The LLM receives documents retrieved by our standard pipeline.

2.3.3 Upper-Bound: Gold Support Paragraph Provided

To simulate an upper bound for evidence quality, we directly provide the model with the gold support paragraph associated with each question. We consider two settings: **oracle gold support**, in which the LLM receives the full gold support paragraph, and **only answer sentence**, in which only the answer-containing sentence is provided.

2.3.4 Lower-Bound: Degraded Evidence Condition

To simulate a lower-bound style condition, we intentionally degrade the quality of the retrieved evidence. We consider three interventions: **remove answer sentence**, where the answer-containing sentence is removed; **add distractor context**, where an irrelevant paragraph is added; and **shuffle context order**, where the order of retrieved contexts is randomized.

2.4 Secondary Experiment Design: Ablation Studies

To understand the impact of practical system design choices, we also performed ablation studies on the Normal RAG pipeline. We varied one of the following parameters at a time while keeping the others fixed:

- **Chunk Size**: 256, 512, and 1024 tokens.
- **Top-k Retrieval**: k = 1, 3, and 5.
- **Embedding Model**: all-MiniLM-L6-v2 and msmarco-MiniLM-L6-v3.

2.5 Evaluation methods

We use the Accuracy as the main metric, measured as the percentage of questions answered correctly. (Lensu, 2025) And we also report recall@k as the secondary metric.(Bhat et al., 2025)

3 Results

3.1 Evidence Quality Study

The RAG baseline setting achieved a 0.81 accuracy and after Oracle gold support provided the accuracy increased to 0.87 (+0.06). In comparison, removing the answer sentence caused a significant performance drop, reducing the accuracy to 0.48 (-0.33), which is close to the LLM-only setting. Adding distractor context resulted a slight decrease in accuracy, while shuffling the context order had a negligible effect on accuracy (0.82). Interestingly, using only the answer sentence also maintained near-baseline performance (0.82), indicating that the exact supporting evidence matters much more than context length or ordering.

3.2 Ablations Study

Ablation experiments demonstrate that the optimal configuration for the Normal RAG pipeline uses a chunk size of 256 tokens and a retrieval volume of k = 3, achieving a maximum accuracy of 0.84 when using the all-MiniLM-L6-v2 embedding model. Although increasing top-k from 1 to 3 significantly improves accuracy, further increasing it to k = 5 does not produce additional accuracy gains and instead increases latency. This indicates that retrieving too many chunks introduces destructive noise into the context window.

4 Conclusion

This study shows that RAG substantially improves SciQ multiple-choice question answering over an LLM-only baseline. The best configuration used a chunk size of 256, top-k = 3, and the all-MiniLM-L6-v2 embedding model. We also find that evidence quality is more important than context length or order: oracle gold support improves performance, removing the answer sentence causes the largest drop, and using only the answer sentence remains close to the baseline. Overall, RAG performance depends mainly on the presence of precise supporting evidence.

References

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A Appendix

Setting	**Accuracy**	**Δ vs. baseline**
LLM-only	0.4373	-0.3720
RAG baseline	0.8093	—
Oracle gold support	0.8727	0.0634
Remove answer sentence	0.4823	-0.3270
Only answer sentence	0.8187	0.0094
Add distractor context	0.7660	-0.0433
Shuffle context order	0.8187	0.0094

Table 1: Results under different settings

Embedding model	**Chunk size**	**Top-k**	**Accuracy**	**Recall@k**	**E2E latency (ms)**
LLM-only	–	–	0.450	–	34.7
all-MiniLM-L6-v2	256	1	0.765	0.485	55.3
all-MiniLM-L6-v2	256	3	**0.840**	0.610	91.9
all-MiniLM-L6-v2	256	5	0.830	0.680	148.9
all-MiniLM-L6-v2	512	1	0.760	0.485	48.9
all-MiniLM-L6-v2	512	3	0.820	0.615	93.6
all-MiniLM-L6-v2	512	5	0.805	0.680	152.0
all-MiniLM-L6-v2	1024	1	0.760	0.485	48.3
all-MiniLM-L6-v2	1024	3	0.820	0.615	93.6
all-MiniLM-L6-v2	1024	5	0.805	0.680	153.8
msmarco-MiniLM-L6-v3	256	1	0.790	0.535	60.1
msmarco-MiniLM-L6-v3	256	3	0.830	0.660	91.7
msmarco-MiniLM-L6-v3	256	5	0.830	0.675	160.0
msmarco-MiniLM-L6-v3	512	1	0.785	0.525	48.5
msmarco-MiniLM-L6-v3	512	3	0.815	0.655	90.7
msmarco-MiniLM-L6-v3	512	5	0.795	0.670	161.6
msmarco-MiniLM-L6-v3	1024	1	0.785	0.525	52.4
msmarco-MiniLM-L6-v3	1024	3	0.815	0.655	92.0
msmarco-MiniLM-L6-v3	1024	5	0.795	0.670	160.3

Table 2: Results of the retrieval ablation study